Gustavo Pinho Maia

PhD Researcher in Chemistry · Astrobiology

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RESEARCH INTERESTS

- **Shock and impact processing.** From planetesimal accretion to asteroid gardening by micrometeorite and IDP impacts, and meteoroid atmospheric entry, extraterrestrial organic matter is repeatedly exposed to mechanical energy.
- **Solid-state mechanochemistry.** Ball milling and grinding of organic analogues (hexamethylenetetramine, ribonucleosides) with minerals and ions found in meteorites (clays, Fe, Ni, Al, carbonate); products characterised by chromatography–mass spectrometry, NMR and DFT calculations.
- **Exogenous delivery and prebiotic chemistry.** Impact processing can both transform and protect organic matter, from forming new N-heterocycles to degrading ribonucleosides into nucleobases, shaping the organic inventory delivered to the early Earth.

EDUCATION

2023 – 2027

PhD in Chemistry (Astrobiology)

Ongoing · Universidade de Lisboa · Instituto Superior Técnico

Mechanochemical energy and prebiotic synthesis

My PhD research focuses on understanding how mechanochemical energy (the energy generated from mechanical forces such as grinding, milling, and impact) can drive the synthesis of prebiotic organic molecules. This work explores alternative pathways for the origin of life that don't rely solely on aqueous chemistry.

2020 – 2022

MSc in Chemistry

Concluded · Instituto Superior Técnico

Ribonucleoside stability and prebiotic synthesis

Exogenous delivery of organic molecules could have played a significant role in prebiotic synthesis, based on the detection of RNA nucleobases and ribose in meteorites. However, ribonucleosides were never detected. Meteoroid atmospheric entry conditions are severe and could lead to chemical modifications via mechanochemical reactions. Moreover, oxidation reactions are also plausible, producing reactive species such as carbonate (CO_3^{2-}).

2017 – 2020

BSc in Industrial Chemistry

Universidade da Beira Interior

Optimization of quinoline N-oxide synthesis

Heterocyclic compounds have been studied since the last century. Due to the presence of heteroatoms, these compounds exhibit unique chemical and physical properties, which allow them to interact efficiently with other molecules or even with biological systems.

2013 – 2016

Specialization Level IV as a Mechatronics Technician

Concluded · Escola Profissional de Espinho

Fully automated robot for Rescue Line Competition (RoboCupJunior)

This program focused on advanced mechatronics and robotics skills. The main project was the design and construction of a fully autonomous robot capable of following a line and performing tasks in the Rescue Line competition at RoboCupJunior.

RESEARCH & PROFESSIONAL EXPERIENCE

Jan 2025 – Present

Visiting Scientist

MNHN · IMPMC, France

Collaborative research at MNHN focusing on mechanochemical synthesis, sample analysis, and astrobiology methods.

Nov 2024 – Dec 2024

Visiting Scientist

NASA Goddard Space Flight Center, USA

This study investigated the influence of mechanochemical processes on the extraction and recovery of organic compounds from extraterrestrial materials. Controlled mechanical activation techniques were applied to evaluate their effects on extraction efficiency, molecular integrity, and degradation pathways.

Sep 2024 – Oct 2024

Invited Assistant (University Teacher)

Instituto Superior Técnico, Portugal

As a Guest Teaching Assistant, I delivered and supervised practical laboratory sessions in General Chemistry for first-year undergraduate students in Aerospace Engineering. The practical lectures were designed to reinforce core chemical concepts through hands-on experimentation, with a strong emphasis on laboratory safety, experimental rigour, and data interpretation.

Jun 2021 – May 2022

President of the Fiscal Council

ANEQ, Portugal

Oversaw fiscal compliance, audited financial statements, and coordinated council meetings.

Jul 2020 – Sep 2020

Research Intern

Universidade da Beira Interior

Conducted laboratory research on chemical synthesis under supervision of faculty researchers.

Jun 2020 – Jun 2021

Vice President

ANEQ, Portugal

Assisted the president and managed organizational activities for student chemistry association.

Oct 2019 – Sep 2020

Delegate of Industrial Chemistry Students

Universidade da Beira Interior

Represented the student body, conveyed concerns, and organized class activities and discussions.

Sep 2019 – Dec 2020

Vice-President, UBIQUÍMICA

University of Beira Interior

Managed student projects, coordinated events, and facilitated communication between members.

Aug 2019 – Sep 2019

Factory Worker

Nestlé Portugal

Aug 2018 – Aug 2019

Scientific Committee Member, UBIQUÍMICA

University of Beira Interior

Jun 2016 – Jul 2016

Participant, RoboCup 2016 World Championship (Junior Rescue League)

World robotics championship · Leipzig, Germany

May 2016

Participant, XVI RoboCup Portugal Open

National robotics competition · Bragança, Portugal

2015

Participant, XV RoboCup Portugal Open

National robotics competition · Vila Real, Portugal

Jun 2014 – Jul 2014

Professional Internship

Ramada Aços, S.A.

May 2014

Participant, XIV RoboCup Portugal Open

National robotics competition · Espinho, Portugal

SUPERVISORS

- **José Armando L. da Silva** · Doctoral advisor · Centro de Química Estrutural · IST
- **Laurent Remusat** · Research supervisor · IMPMC · Muséum national d'Histoire naturelle
- **Jason P. Dworkin** · Research supervisor · NASA · NASA Goddard Space Flight Center

PEER-REVIEWED PUBLICATIONS

1. **Gustavo P. Maia** et al. (2026). **Effect of asteroid gardening on the exogenous delivery of extraterrestrial organic matter**. *Earth and Planetary Science Letters*. https://doi.org/10.1016/j.epsl.2026.120237
2. **Gustavo P. Maia**, Catarina Gonçalves, Ana J. Carvalho, Vânia André, Adelino Galvão, Ana P. C. Ribeiro, Pedro F. Pinheiro and José Armando Luísa da Silva (2025). **Mechanochemical Reactivity of Ribonucleosides Mediated by Inorganic Species: Implications for Extraterrestrial Organic Matter Interpretation**. *Applied Sciences*. https://doi.org/10.3390/app15031363
3. **Gustavo P. Maia**, José Armando Luísa da Silva, Vânia André and Adelino M. Galvão (2023). **Shock-Induced Degradation of Guanosine and Uridine Promoted by Nickel and Carbonate: Potential Applications**. *Molecules*. https://doi.org/10.3390/molecules28248006
4. Ana Franco, **Gustavo P. Maia** and José Armando Luísa da Silva (2023). **Da Química Bioinorgânica para a Química Prebiótica: Levar Mais Longe o Legado do Prof. Fraústo da Silva**. *Boletim da Sociedade Portuguesa de Química*. https://doi.org/10.52590/M3.P708.A30002745

CONFERENCE PRESENTATIONS

Aug 2026
Effect of Mechanochemistry on Organic Matter: Implications for Impact Gardening on Asteroid Surfaces ORAL
MetSoc 2026 · Frankfurt
Oral talk · Maia, Remusat, Amano, Viennet, Dworkin, McLain & da Silva

Oct 2025
The effect of mechanochemistry on the formation of extraterrestrial organic matter ORAL
AbGradE'25 · Lisbon
Oral talk · Maia, Dworkin, da Silva & Remusat

Jul 2025
Effect of mechanochemical events on extraterrestrial organic matter and its implications for exogenous delivery ORAL
BEACON 2025 · Harpa, Reykjavik
Oral presentation · BEACON 2025

Jun 2025
The effect of shock/impacts on the exogenous delivery of organic matter ORAL
IMPMC PhD Day · Paris
Oral talk · IMPMC, Sorbonne Université

Jan 2025
Mechanochemistry of extraterrestrial bodies: Bridging exogenous delivery with prebiotic chemistry ORAL
MNHN · Paris
Oral talk · MNHN, Paris

Jan 2025
Mechanochemistry and astrobiology: mimicking impact reactions ORAL
MNHN · Paris
Oral talk · MNHN, Paris

2025
Solid-state mechanochemistry of extraterrestrial organic molecules: new insights on the exogenous delivery of organic matter POSTER
EANA 2025 · Lisbon · Poster Award
Award-winning poster · EANA 2025

2025
Mechanochemistry of small bodies and the evolution of extraterrestrial organic matter ORAL
IPGP "Small Bodies Day" 2025 · Paris
Oral talk · IPGP, Paris

2024
Impact events of extraterrestrial bodies: the bridge between exogenous delivery and the origin of life ORAL
NInTec Science Days 2024 · Lisbon
Oral talk · NInTec Science Days 2024

2024 **Shock-Impact Events and the Origin of Life: a mechanochemical approach** POSTER
IST PhD Open Days 2024 · Lisbon
Poster · PhD Open Days 2024

Aug 2022 – Sep 2022 **Ribonucleoside stability and prebiotic synthesis** ORAL
8th EuChemS Chemistry Congress · CCL, Lisbon
Oral talk · 8th EuChemS Congress

2022 **Ribonucleosides stability and prebiotic synthesis** POSTER
CQE Days 2022 · Lisbon · P083
Poster · CQE Days 2022

AWARDS & DISTINCTIONS

2025 **EANA 2025 Poster Award**
EANA 2025 · Lisbon, Portugal
Poster Award at the European Astrobiology Network Association annual conference, recognising research on mechanochemistry and the prebiotic synthesis of organic molecules.

2017 **Diploma of Merit Recognition**
Rotary Club de Espinho
Merit recognition diploma awarded for outstanding academic performance in 2015/2016 at the Escola Profissional de Espinho (CEPROF).

2016 **Best Presentation · RCJ Rescue Maze**
RoboCup 2016 · Leipzig, Germany
Awarded to team ROB07s (Escola Profissional de Espinho) at the RoboCup world championship for the best presentation in the RoboCup Junior Rescue Maze league.

FUNDING

2023 **Mechanochemical energy and prebiotic synthesis**
Doctoral Grant · Fundação para a Ciência e a Tecnologia (FCT) · 2023.01099.BD
DOI: 10.54499/2023.01099.BD

Jun 2023 – Nov 2023 **A Gold Nanosensor for Single Molecule Detection of RNA Biomarkers in a Miniaturized Microscope**
Research Fellow · Centro de Química Estrutural, Universidade de Lisboa

2020 **Centro de Química Estrutural**
Research Unit Funding · Fundação para a Ciência e a Tecnologia (FCT) · UIDB/00100/2020
DOI: 10.54499/UIDB/00100/2020

COURSES & TRAINING

Jun 2026 **Building the Hard Rocky Planets — From Mercury to the Asteroid Belt**
Summer School · Institut Origines (Aix-Marseille Université) & Institut de Physique du Globe de Paris (IPGP) · Marseille, France

Jun 2026 **Astrobiology Introductory Course (Virtual)**
Course · Rencontres Exobiologiques pour Doctorants (RED) · Le Teich, France · Online

Mar 2025 **Astrobiology Introductory Course**
Course · Rencontres Exobiologiques pour Doctorants (RED) · Le Teich, France

Feb 2025 **Recommendations for Publishing in International Journals**
Formation · Biblioteca do Instituto Superior Técnico (BIST) · Online

TECHNIQUES & SOFTWARE

Analytical techniques	Chromatography & MS, NMR, XRD, FTIR, EA, irMS, SEM · EDX
Instruments & data	TopSpin, MestReNova, OpenLab, MassLynx & TargetLynx, OPUS, Profex, ACD/Labs
Synthesis & modelling	Mechanochemistry, Organic Synthesis, Ab Initio

Chemistry & computation	Python, R, MATLAB, ChemDraw, Chemcraft
Engineering & electronics	Arduino, PICAXE, Multisim, SOLIDWORKS, AutoCAD, SketchUp
Productivity & code	Office 365, VS Code, GitHub

MEDIA & RECOGNITION

2026	Recognised among the “Excellent Teachers 2024/2025” at Instituto Superior Técnico (Universidade de Lisboa), for teaching in the General Chemistry laboratories.
Sep 2025	Featured in Maré Viva (Jornal Regional de Espinho): “O percurso improvável de Gustavo Maia levou-o de Ovar até à NASA”, a profile tracing the path from Ovar and the Escola Profissional de Espinho, through world robotics competitions, to a NASA laboratory.
Dec 2024	Interviewed by Praça Pública: “Gustavo Maia aprofunda investigação no Laboratório Analítico de Astrobiologia da NASA”.
Nov 2024	Interviewed by Diferencial, the IST student journal: “Portanto, a Vida foi criada a partir de surras.”

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